# Historical Britannica: extracted source

Encyclopaedia Britannica, or, a Dictionary of arts, sciences, and miscellaneous literature : enlarged and improved. Illustrated with nearly six hundred engravings - Fifth edition, Volume 17, Poetry-RHI - EB.10

Source row 386. Parser: nvidia/nemotron-3.5-parse.

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Region region-0; provider order 0.



Region region-1; provider order 1.

have committed great milhkes by not attending to this; and it is necesfary to notice if jul many, bequels calls will occur in the protection of this fubjeb, where we shall be very apt to confound our realonings by a confusion in the ufe of those reactions. These mathematics who are secondoured to the geometrical consideration of cavrili-neal motions, are generally diploed to take the _defined de_- _filers than the tangent as the measured of the deflecting force: ; while while what treat the fun fulgled algestralically, by the afflance of flusions, _take_ _the change of velocity_, which is mediated by reader the definition. The reason to this, when a body suffer through the point B of a curve ABC, fig. 1. iF the deflecting force were to cease in that nation, the body would describe the agent BD in the time but in which it describes the arch BC of the curve, and DC is the deflection, and is therefore taken for the mediums of the defuding force. But the algebraic is continuous to onder the curve by means of an equation between the abdiffe H, A, H, F, and their respective estimates for , B, K C, e, and he measures the deflection by the changes made on the increments of the ordinates. Thus the increment of the ordinary A, or while the body describes the arch AB of the curve, is BG. If the deflections force were to cease when the body is at B, the next increment would have been equal to BG, that is, it would be seen SIF; but in consequence of the deflection, his only CW there– face the takes EC for the measure of the deflection, and of the deflecting force. Now DO is ultimately one DC, and thus the measure of the algebraic (derived daily from the nature of the differential method, and without any regard to physical conditions) happens to coin- cide with the ten payable markers. There is therefore

Region region-2; provider order 2.

Particularly great danger of making these meures. Of this we can **of**Libiniz not give a more remarkable influence than Leibniz's attempt to demonstrate the elliptical model of the planets in the Leipde Acts, 168q. He firft considers the bab-jedi mechanically, and then the deflection or DC for the medium of the defuding force. He then nito-duces his differential calculus, where he takes the difference of the increments for the meanness, and thus brings himself into a condition, which quickly compete for the fallle reasoning in the preceding part of his paper, and gives his mind the appearance of a demonstration of Newton's great discovery, while, in fact, it is a confudet jumble of affirmations, left-con- tradictory, and inconflient until the very laws of me sharing which are ruled by him in the investigation. Seventeen years after this, in 1750, having been critied for the had resolving or rather accused of an envious and unificational attempt to appropriate Newton's invention to himself, he gives a correction of his paradingin, which has calls a correction of language. But he either had no delivered where the programing by, or would not let himself down by acknowledging a mikaks in what he waffed the world to think his own calculus (fluxion) , he helped the correlation where the fault had been committed, for he had mentioned both the centrifugal force and the distinction of gravity in the way, but had applied the fluxionisme experiment on the hill and wait to the lift, and, by fo doing, he com-pletely destroyed all coincidence between his critical the planetary motions. We mention this influence, not only as a curious to our mathematical readers, but alfo as a very curious literary anecdote. This dissertation of

Region region-3; provider order 3.

Leibniz is one of the most obscure of his oblicure writ-ings, but deferred the attention of an intelligent and curious reader, and cannot fail of making an indelible impectition on his mind, with relation to the modeller, curdours and posibly of the author. It is preceded by a diffraction on the fulgbelt which we are now entering upon, the modest properties in writing medium. Newton _Principle_ and been published a few years before, and had been reviewed in a manner flamo- fully flight, in the Leipde Acts. Both the fulgbelt make the capital articles of the immense work. Mr Leibniz published the differifications, without (fays he) having few Newton's book, in order to flow the world that he had, the years before different the time theorems. Mr Nicholas Patio carried a copy of the _Principe_ from the author to Hanover in 1686, where he expected to find that Leibniz's law and then defendant, but Patio faw him often before his return to France in 1687, and does say that the book was not given him. Read along with the description of KeRIS letters to John Bernoulli and others, published in the _Journal Litic- rare de la Haque_ 1714, and to John Bernoulli in 1700.

Region region-4; provider order 4.

Newton has been cancelled, a limited averagely by John Bernoulli, (who indeed calls it a mitake in princided of a cipelo) in his Proposition x book z on the very lab-mikas jech we are now considering. But DrKell has shown by Jt to be only an overflight, in drawing the tangent on the wrong side of the ordinate. For in this very pro-pollution Newton exhibits, in the flatfield must bear- find manner, the difference between the geometrical and algebraic manner of confining the fulgbet; and expertly wrote the reader, that his spherical symbol expresifies the deflection only, and not the variation of the inherent of the ordinate. It is therefore in the Pad failed lat degree impossible that he would make the miif- take. He molt expressly does not; and as to the real mitke, which he corrected in the second edition, the writer of this article has in his pdfiction a manuscript copy of name and illustrations on the whole Principia, written in 1692). DrD and Gregory Savlan pointed off of alternony at Oxford, at the die of Mr Newton, as preparatory for a new edition, where he has rectified this and several other mistakes in that work, and says that Mr Newton had dem and approved of the amendments. We mention thefe particulars, because Mr Ralucety Bernoulli published an elegant diffraction on this dataset fulgbelt in the Leipde Acts in 1713; in which he with re charges Newton (though with many protefations of speed) admiration and respect] with this mistake to principle, Nden and fays, that he commented his correction to Mr Newton, by his nephew Nicholas Bernoulli, that it might be corrected in an new edition, which he heard was in the prefix. And he afterwards adds, that it ap-pears by fome cheats being cancelled, and new ones fab- filtered in this past of the work, that the mitke would have continued, had the not correct how it. We would define our readers to confult this differentiation, which is extremely elegant, and will be of review to us in this article, and let them compare the end! things which is here faid of the vir incomporabilis, the omni laude major, the _fammina Neuenus_, with what the fine author, in the time years, in the Leipde Acts, the under a bor- rowed name, fays of him. Our readers will have no hesitation in afcribing this letter to this author. For, after printing John Bernoulli au _fammut gementos_.

Region region-5; provider order 5.

## Source record

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